

Computing and Software

SFWRENG 2XC3

Software Engineering Practice and Experience: Development Basics

Fall 2024

Instructor Information

Samuel Scott

Email: scotts52@mcmaster.ca

Office: ETB 111

Office Hours:

Fridays 10am to 12pm

Feel free to email any time. If you are working on an exercise or assignment, you can send your code along with your questions or a description of the problem you're having, and I'll try to steer you in the right direction.

Class Times

Friday, 3:30pm to 4:30pm, BSB B136

Class Format

In Person

Each course week will start with a short lecture that introduces important topics. The following two 2-hour labs will fill in details, give you practice exercises, and allow time for one-on-one help with your TA. Labs will also contain activities to be submitted for a small completion grade.

The course begins with the first lecture (labs scheduled prior to the first lecture are cancelled).

Attendance and in-person participation in classroom activities is strongly recommended.

Course Dates: 09/03/2024 - 12/05/2024

Units: 3.00

Course Delivery Mode: In Person

Course Description: Imperative low-level imperative programming, with documentation, testing, benchmarking, profiling and debugging, revision control. Acquiring professional software development skills via practical experience with interaction with UNIX-like systems, shell interaction and programming, pipes and filters. One lecture, two labs (two hours each); first term.

Prerequisite(s): ENGINEER 1D04 or 1P13 or IBEHS 1P10; registration in any Software Engineering

program Antirequisite(s): COMPENG 2SH4, COMPSCI 1XC3, 2XA3, 2S03, MECHTRON 2MP3 , SFWRENG 2MP3, 2S03, 2XA3

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [C Programming: A Modern Approach](#)
- [The Linux Command Line](#)
- [Official Git Documentation](#)
- [Equity and Inclusion](#)
- [2SLGBTQIA+ Resources](#)
- [Black Lives Matter](#)
- [Indigenous Student Services](#)
- [MES Clubs](#)

Course Learning Outcomes

1. Navigate files and directories from the Linux command line.
2. Solve information extraction tasks on the Linux command line using pipes, filters, and redirection.
3. Plan and manage individual and group software development projects using version control with Git and GitHub.

4. Implement a continuous integration pipeline on GitHub including automated build, test, analysis, and reporting steps.
5. Build command-line filter apps in both bash and C that take input from standard in, process it according to the options and option arguments provided, and send the output to stdout.
6. Solve C programming problems involving basic data types, arrays, strings, structures, pointers, and the standard input, output, and error streams.
7. Optimize C code using performance analysis tools, code refactoring, and preprocessor directives.
8. Manage dynamic memory allocation and deallocation effectively in a C program.
9. Develop high quality, error-free C code using unit, integration, and regression testing along with profiling tools for code coverage tools and performance .

Graduate Attributes

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national standard of quality and cover essential educational requirements. Graduate Attributes are a set of qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

- Graduate Attribute Indicator 1.4: Competence in Specialized Engineering knowledge. (Course Learning Outcome 6)
- Graduate Attribute Indicator 2.1: Identifies and states reasonable assumptions and suitable engineering fundamentals, before proposing a solution path to a problem. (Course Learning Outcome 8)
- Graduate Attribute Indicator 3.2: Synthesizes the results of an investigation to reach valid conclusions. (Course Learning Outcome 7)
- Graduate Attribute Indicator 4.1: Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions. (Course Learning Outcome 6)
- Graduate Attribute Indicator 11.1: Applies economic principles in decision making. (Course Learning Outcome 3)

- Graduate Attribute Indicator 11.2: Plans and effectively manages a project's time, resources, and scope, following business practices as appropriate. (Course Learning Outcome 3)

Lab Information

- **L01:** Tuesdays and Thursdays, 12:30 - 14:20, BSB 244
- **L02:** Mondays and Wednesdays, 12:30 - 14:20, BSB 244
- **L03:** Mondays and Wednesdays, 12:30 - 14:20, ETB 118
- **L04:** Tuesdays and Fridays, 16:30 - 18:20, KTH B123

Course Schedule

The Course Week

Each course week starts with a lecture and includes the two labs that follow that lecture. Note that the class begins with the first lecture of the semester. Labs scheduled before the first lecture are cancelled. Due to this semester's schedule, there are 12 course weeks in total.

Legend:

- **C:** Readings from *C Programming, A Modern Approach*.
- **L:** Readings from *The Linux Command Line*.
- **G:** Readings from the official Git documentation.
- **A1-A4:** Individual assignments. **A5:** Team assignment.
- Weekly lab submissions (5% total) not shown

		Reading	Topics	Major Evaluations
1	Sep 6 - 12	C: 1, 2.1-2.3 L: 1-5, 12	The CI Pipeline: Continuous integration, the Linux environment, Linux file systems, the C environment.	

2	Sep 13 - 19	C: 5.1-5.2 G: Git Basics, Pro Git 1-2	The CI Pipeline: Version control, Git and GitHub, workflows, a little bit more C.	A1 Begins
3	Sep 20 - 26	C: 2.4-2.8, 3, 6.1, 6.4	Linux: Standard input/output/error, redirection, here docs and strings, pipes and filters, bash scripting basics, creating new filters in C and bash.	
4	Sep 27 - 3	L: 19, pp. 402-407	Linux: Regular expressions, effective use of grep, more bash scripting.	A1 Due (Oct 3, CI Pipeline, 5%) A2 Begins
5	Oct 4 - 10	C: 4, 5.3, 6.2-6.3,7	Programming in C: preprocessor, compiler, linker, type system, expressions, control structures.	The Linux and Git Test (20%)
6	Oct 11 – 24 (break week)	C: 9 (skip pp. 195-201), 11.1-11.4, 15	Programming in C: Functions, program structure, pointers, recursion, header files, makefiles, unit testing, coverage analysis.	A2 Due (Oct 24, Bash, 10%) A3 Begins
7	Oct 25 - 31	C: 8, 12, pp. 195-201	Programming in C: Arrays, pointers, pointer arithmetic	

8	Nov 1 - 7	C: 17.1, 17.3, 17.4	Programming in C: Dynamic array allocation, memory management	A3 Due (Nov 7, C Basics, 5%) A4 Begins A5 Teams Formed (1%)
9	Nov 8 - 14	C: 14.1, 14.2, 14.3 (pp. 334-335) G: Pro Git 3	Advanced Git: Branching, merging, branching strategies Programming in C: Performance analysis, optimization, preprocessor directives, macros	A5 Plan Due (4%)
10	Nov 15 - 21	C: 13, 17.2	Programming in C: Strings	A4 Due (Nov 21, C Arrays, 10%)
11	Nov 22 - 28	C: 16.1-16.3, 17.5-17.6	Programming in C: Structs and linked lists	A5 Increment Due (2%)
12	Nov 29 - 5	G: Pro Git 5-6	Advanced Git: Collaboration and collaborative workflows, review	A5 Final Due (8%)
EXAM WEEK				The C Test (30%)

Required Materials and Texts

Textbook Listing: <https://textbooks.mcmaster.ca>

C Programming: A Modern Approach, Second Edition

Authors: K. N. King

Publisher: Norton & Company

Publication Date: 2008

Edition: Second

The Linux Command Line

Authors: William E. Shotts Jr.

Publisher: <https://linuxcommand.org/tlcl.php>

Publication Date: 2019

Pro Git

Authors: Scott Chacon and Ben Straub

Publisher: <https://git-scm.com/doc>

Official Git Documentation

Publisher: <https://git-scm.com/doc>

Handouts, Slides, Examples, Readings, and Class Notes

Available through Avenue to Learn

Course Evaluation

- Linux and Git Test: 20%
- C Test: 30%
- Lab Milestones (~1 per week): 5%
- Individual Assignments (4): 30%
- Team Assignment: 15%

Course Evaluation Details

Tests

Tests are worth 50% in total. They will be closed-book, though you may be provided with reminder sheets for the details of common Linux, Git, and/or C commands and syntax. The first test will mostly cover the basics of Linux and Git and will be written during lab time in week 5. The second test will mostly cover C programming and will be written during a scheduled final exam session after the end of classes.

If you miss a test for any reason, or use an MSAF for a test, you will still have to write it. You should contact the professor as soon as possible to reschedule the test.

Labs

Most lab meetings will have one or more items to hand in. These are short exercises or activities that are intended to be completed during lab time. They will be due at the end of each course week. They will be graded as complete or incomplete (0 or 1). They are all equally weighted.

If you choose to MSAF one of these activities, you will be given a grade of "complete" for that submission.

Assignments

The assignments will involve writing apps and then automating the building, testing, and/or reporting steps of a Continuous Integration pipeline using Git and GitHub. You are encouraged to discuss the course material with your classmates, but the four individual assignments must be your own individual effort.

Every assignment and exercise in this course is carefully curated so that you should not have to go outside the course materials and assigned readings to find the tools and techniques required. When figuring out how to do an assignment, you should always go to the course materials and readings first. You may consult other sources for short code snippets, but all such sources must be documented. Failure to do so may be a violation of academic integrity.

The use of ChatGPT and other Large Language Models (LLMs) is strongly discouraged during any part of the assignment completion process. In this introductory course, it is crucial that you take every opportunity to write your own code and solve programming problems yourself. This is the only

way to develop the skills you will need to use LLMs effectively in the future for more complex and specialized software development. Researching specific issues online and discussing solutions with classmates can aid your learning, but the use of LLMs in an introductory programming course can short-circuit the learning process. Using code produced by ChatGPT or any other Large Language Model is considered a violation of academic integrity.

See below for the late and MSAF policies.

Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Late Assignments

Late Individual Work

The individual assignments are due on the day listed in Avenue (i.e., up to 11:59:59 pm that day). Lab activities are due at the end of the course week in which they were assigned. After that, they are late. Late days are tracked by calendar date. For example, if the due date is November 1st and you hand the assignment in some time on November 2nd, it is one day late.

You can hand in late assignments or labs with no penalty, but you have a limit of 8 late days for all the individual assignments and labs combined. If you go over that limit, late work will no longer be accepted. Please note that the final day to hand in any work for this course, regardless of how many late days you have remaining, will be the final day of classes (December 5th).

Your 8 late days do not include special or unforeseen circumstances (illnesses, family emergencies, etc.). If you are delayed in completing an assignment due to special or unforeseen circumstances, let your professor know as soon as you can so that you can negotiate a new due date. Keep in mind that you may be required to produce supporting documentation.

Late Team Work

For the team assignment, if there is some reason your group is unable to deliver on the due date, someone from the group must contact the professor to discuss the situation.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it “shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source”. This includes work created by generative AI tools. Also stated in the policy is the following, “Contract Cheating is the act of “outsourcing of student work to third parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/), located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer

activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Equity, Diversity, and Inclusion

The Faculty of Engineering is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, in particular if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or to contact the [Equity and Inclusion Office](#).

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a

Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Academic Advising

For any academic inquiries please reach out to the Office of the Associate Dean (Academic) in Engineering located in JHE-Hatch 301.

Details on academic supports and contact information are available from:

<https://www.eng.mcmaster.ca/programs/academic-advising>

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations.

Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law

protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.